# Use work-energy to solve motion

**antocciphysics**

UNIT 4 / DAY 5 / HONORS

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Choose work-energy when force and displacement are known.

**SUCCESS CRITERIA - I can...**

1. List all forces doing work.
2. Calculate the kinetic-energy change.
3. Check whether the result is physically possible.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

A force parallel to motion does 15 J over 3 m. Find its size. What work does the same force do with zero displacement?

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $W_{net} = \frac{1}{2}m(v_f^2 - v_i^2)$.
Constant mass. Use the displacement along each force component; stopping removes the initial kinetic energy.

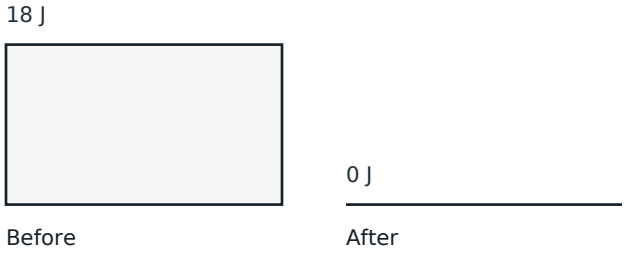
**Worked reasoning:** A 2 kg cart starts from rest. Constant net force 4 N acts through 3 m, doing 12 J. Then $\frac{1}{2}mv^2=12$ gives $v=\sqrt{12}\approx3.46$ m/s.

**Your turn:** Sketch a force diagram and an energy bar diagram for this motion. Identify where time is unnecessary.

**Marzano check - circle:** ① ② ③

## 03 Represent and interpret

A cart has 18 J kinetic energy and then stops. How much net work is done? Show an energy bar before and after.




Bar height represents energy. Read the value and unit above each bar.

**Marzano check - circle:** ① ② ③

## 04 Solve and justify

A 2 kg cart moving at 6 m/s encounters a constant 3 N resistive force. Find stopping distance and compare with the distance if initial speed doubles.

**Marzano check - circle:** ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

### 05 Analyze the reasoning

A calculation gives negative final kinetic energy after a long braking distance. What should you check?

**Marzano check - circle:** 1 2 3

### 06 Choose a method

Initial momentum is 0.4 kg·m/s and measured final momentum is 0.38 kg·m/s. Find percent deviation. Explain why this is not itself a measured uncertainty.

**Marzano check - circle:** 1 2 3

### 07 Exit check

A 5 kg cart starts at rest. A 12 N pull and 2 N resistance act over 4 m. Find final speed.

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

A 4 kg cart at 5 m/s slows to 1 m/s over 6 m. Find the constant net force component along its motion.

**Marzano check - circle:** 1 2 3